

Batch Gradient Ascent — Cauchy MLE — Exam-Ready Short Answers

Q1. Goal of the algorithm?

Find MLE μ -hat maximizing Cauchy log-likelihood $\ell(\mu)$ via gradient ascent (fixed step), not Newton-Raphson.

Q2. Data setup?

$n=4$, true $\mu^*=5$, $X = \text{rt}(4, \text{df}=1) + 5$ (t_1 shift = Cauchy). Tiny $n \rightarrow \ell(\mu)$ can be non-concave.

Q3. Log-likelihood formula?

$\ell(\mu) = -n \cdot \log(\pi) - \sum \log[1 + (x_i - \mu)^2]$.

Q4. Score f.prime and curvature f.double.prime?

$f' = \sum 2(x_i - \mu) / [1 + (x_i - \mu)^2]$. $f'' = \sum 2\{2(x_i - \mu)^2 / [1 + (x_i - \mu)^2]^2 - [1 + (x_i - \mu)^2]^{-1}\}$. f.double.prime is defined but never used in the update.

Q5. Gradient ascent update rule?

$\mu^{(k+1)} = \mu^{(k)} + t \cdot f'(\mu^{(k)})$, $t=0.3$ fixed — simpler than NR, no Hessian, smaller/slower steps.

Q6. Stopping rule?

while($\text{diff} > \text{tol} = 10^{-5}$ AND $\text{iter} < 1000$): $\text{diff} = |\text{update} - \text{current}|$ each step; cap prevents infinite loop.

Q7. Three scenarios run — why?

Same loop, 3 starts (7, median(X), 19) — shows sensitivity to initial value on non-concave $\ell(\mu)$.

Q8. Role of the parabola (approx) in the plot code?

$\text{approx} = \ell(\mu_0) + f'(\mu_0)(\mu - \mu_0) - (\mu - \mu_0)^2 / (2t)$ — local quadratic view at each visited point; illustrative only, not used to compute the step.

Q9. Sys.sleep(2) purpose?

Pauses 2s per plotted point — animates climb in live R session; irrelevant/skippable in a static/knit output.

Q10. Actual numeric results (seed=1)?

- Bad (start=7): 40 iters $\rightarrow \mu\text{-hat} \approx 4.5621$
- Good (start=median ≈ 4.597): 30 iters $\rightarrow \mu\text{-hat} \approx 4.5621$
- Worst (start=19): 23 iters $\rightarrow \mu\text{-hat} \approx 16.1631$ (different point!)

Q11. Why does "worst" not fail/diverge, but still is wrong?

It converges cleanly — to a genuine local max, not global. $\ell(4.562) \approx -9.57 > \ell(16.163) \approx -19.40$. Outlier in X creates 2nd peak; $\mu_0=19$ sits in its basin.

Q12. What does the combined plot show?

$\ell(\mu)$ is bimodal: tall peak near 4.5, short peak near 16–20. Red & blue dots climb tall peak (overlap); green dots climb short peak only — never cross the dip.

Q13. Conclusion?

Fast, clean convergence $\neq$ correctness on non-concave $\ell(\mu)$; starting value can silently select the wrong local optimum.